BLOOM design and working report

From Cherry to Cup · Multimedia Production Implementation and Delivery

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COMP5415/4415 Multimedia Design and Authoring

Assignment 2

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Tutor and class: Alex Hao | Tutorial 02 - Time 8-9 pm

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Figure 1. The floating BLOOM world, a Blender render of Scene 3.

Declaration

I declare that this assignment is my own work, except where I have acknowledged the work of others. All sources, including the music, sound effects and reference images, are credited in the Reference section, and all use of generative AI and other tools is acknowledged in Section 7 and in my AI acknowledgement form.

I have not submitted this work for assessment in any other unit. I understand the University of Sydney's Academic Integrity Policy and that this work may be checked for originality.

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Project overview

BLOOM is a short animated story about how a coffee cherry becomes a cup of coffee. A girl with a wand, a hovering basket and a bee travels across floating islands. She harvests cherries, processes them at a little mill and finishes with a pour into a giant cup, where a blossom opens on the coffee. The story is told in three 40-second episodes, two minutes in total, and is delivered through a local website.

This report follows the Assignment 2 template. It explains my design decisions, the techniques I used, and exactly which parts were my own work and which had AI help. The table below gives the summary, and each section gives the detail.

Who did what at a glance

Part	Tools	My own work	AI assistance
Story and narration script	Written by me	The three-episode story and the narration script	Verified if it is doable in blender.
Logo	Figma	Made the final logo in Figma	Claude helped with ideation only
Posters (3)	Figma, Blender renders	Made the posters in Figma from my own Blender renders, and chose the frames and titles	Claude helped with ideation only
3D modelling and animation	Blender	Built and animated everything in Blender, including the wave in the Graph Editor	ChatGPT gave step-by-step instructions and Python scripts, and helped fix overlaps and errors
Video compositing	Blender compositor	Set up and explored Alpha Over composition out of my own interest	ChatGPT instructions for the compositor setup
Narration audio	My voice, Canva	Recorded the voice myself, then edited and synced it in Canva	ChatGPT was used to alter the pitch of my own voice
Music and sound effects	ChatGPT, Canva	Combined the music and effects with the narration and video myself in Canva	ChatGPT was used to fetch the music and sound effects
Captions	Canva, WebVTT	Wrote the words being captioned and timed each caption to my narration by hand in Canva	None

Part	Tools	My own work	AI assistance
Website	HTML, CSS, JavaScript	Coded the site myself, and supplied the content, media, credits and final checks	Claude assisted with ideation, helped deploy the site on GitHub, and enhanced my basic design with animation and bee graphics
Presentation	Canva	Reworked and finished the slides in Canva	Claude drafted the structure and speaker notes
This report	Word	My account of the work, which this report is based on	An earlier draft was prepared with ChatGPT; this version was rewritten with Claude

Table 1. Who did what at a glance.

1 Theme and Design

1.1 Project goal and target audience

The goal of BLOOM is to show the steps of making coffee as a gentle fantasy adventure. It has three aims: to tell the harvest, roast and grind, and brew stages as three short episodes; to deliver them through a local website that is easy to navigate; and to be open about where AI was used.

The intended audience is visitors who enjoy short visual stories and playful coffee branding. The project gives them a calm, charming film that shows the journey of coffee in an easy-to-follow way. This is design intent, not a tested audience finding. The milk-through-filter sequence is a fantasy device and should not be read as a brewing tutorial.

1.2 Design ideas and design considerations

Central idea. The central idea is transformation: a small fruit becomes a familiar drink through a character journey. Floating islands and oversized equipment make everyday coffee preparation feel like a storybook.

Visual style. Rounded forms, soft shading and miniature proportions give a toy-like world. The girl, wand, hovering basket and bee are recurring points of attention. Red cherries stand out against green leaves, and cream equipment separates the process objects from the scenery. Camera changes alternate wide views for orientation with closer views of actions.

Consistent aesthetics. I kept the aesthetics and colour palette consistent across the film, the logo, the posters, the presentation and the website. The film uses the bright mood-board colours. The logo, posters and website use a darker espresso background with cream, kraft and ember-orange accents taken from the same coffee theme.

Mood board. Before building, I gathered a mood board of reference images: floating islands, a red wagon, a bee, cherry blossom, a coffee cherry cross-section, brewing icons, a castle and a pouring

waterfall of coffee. It set the fairy-tale tone and the cherry-to-cup process that the film follows. The images were taken from Pinterest and Google Images for creative reference only, and nothing more. None of them is used in the final film, posters or website, which are built from my own Blender renders and Figma artwork.



Figure 2. The BLOOM - From Cherry to Cup mood board. Reference images from Pinterest and Google Images, used for creative reference only.

Colour	Hex	Where it is used
Meadow Green	#7BC86C	Landscape and vegetation in the film
Blossom Pink	#FFB7D0	Flowers and soft accents in the film
Cherry Red	#E5484D	Fruit and cart emphasis in the film
Coffee Brown	#6B3F26	Coffee and warm details in the film
Morning Sky	#8FD3F4	Background colour family in the film
Espresso	#1F120C	Website, logo and poster background
Cream	#FBF7F0	Text, logo petals and light surfaces
Kraft	#C9A06A	Borders and secondary text on the website and posters
Ember	#D9772F	Accent: logo centre, buttons and highlights

Table 2. Colour palette. The film colours come from my first-draft mood board (Appendix A) and are references, not a claim that every shader uses the same value.

Name and meaning. The name BLOOM links the flower motif to growth and transformation. The final cup carries a blossom, and the logo is a blossom too, so the end of the story and the brand match.

How the concept developed. The early plan used rolling hills and a farmhand. The implemented version uses floating islands and a girl with a wand, so this report uses the current structure. Scene 3 was first called The Big Pour and is now titled The Bloom, to match the blossom ending.

1.3 High special features

I explored four features beyond the basic lab techniques. Each is explained in more detail in its own section.

- **Bezier curves and procedural animation with Python scripts.** I got the idea from the lecture slides, which taught these concepts, and used it to build the milk stream, the coffee waterfall, hanging roots and stems, and to calculate limb poses by script (Section 3).
- **Alpha Over composition.** I explored separating the girl from the environment into two render layers and joining them with Alpha Over nodes. I did this out of pure interest, beyond what was required (Section 4).
- **A hand-animated wave.** I first tried FreeMoCap for the girl's wave, but my laptop could not process it, so I keyframed the wave in Blender's Animation and Graph Editor workspaces (Section 3.5).

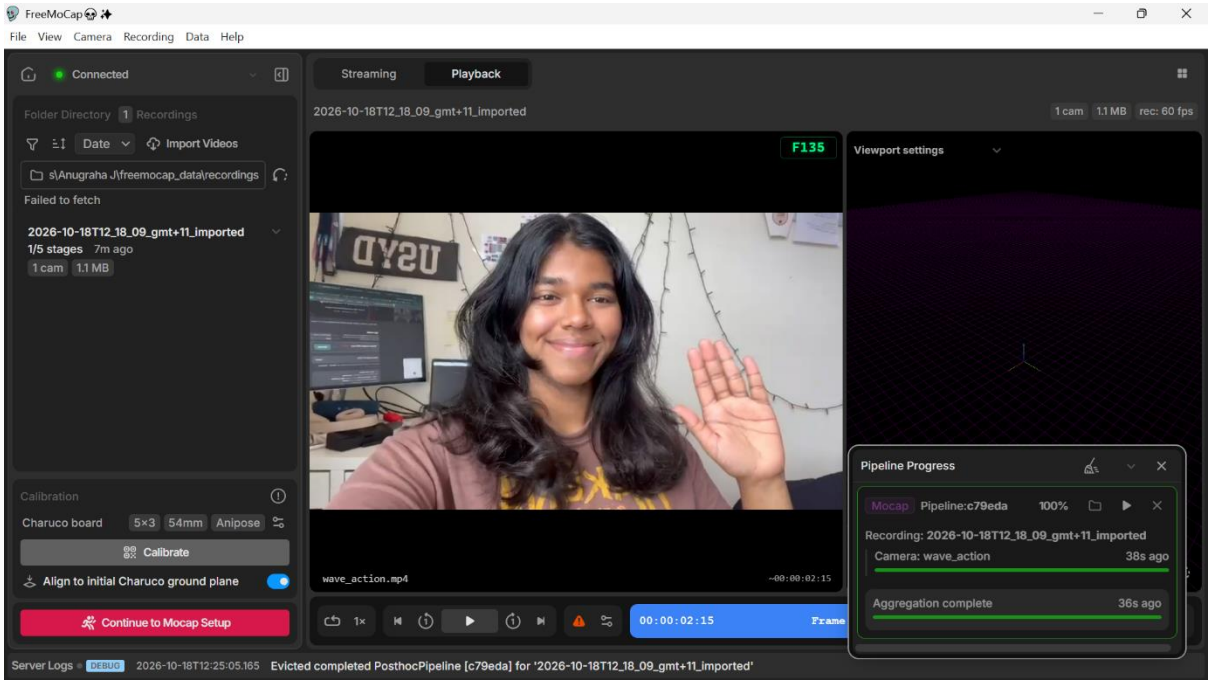


Figure 3. Recording the wave reference in FreeMoCap, my first attempt at motion capture for the girl's wave.

- **A credited, accessible website.** The website has captions, keyboard access, and a page that credits both the AI use and the tools used (Section 6).

1.4 Story board

First-draft storyboard and mood board

My first draft was a one-page mood board and shot-by-shot storyboard for a two-minute cartoon film, ideated with Claude in chat and designed in Figma by me. The full page is in Appendix A and its key frames are below.

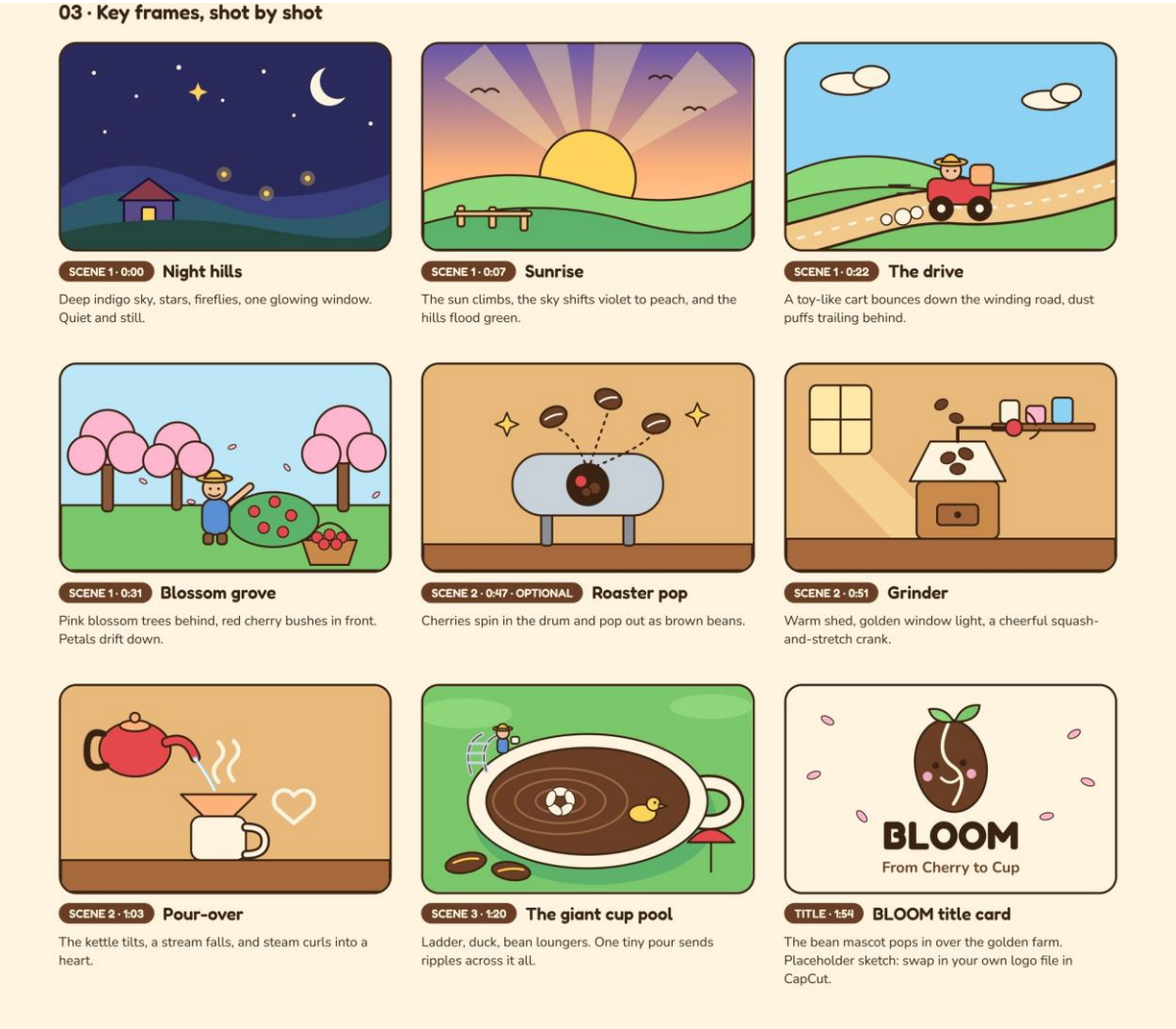


Figure 4. Key frames from the first-draft storyboard, shot by shot (detail of Appendix A).

The idea then changed to a fairytale land as I had in my mood board and built it in Blender. The table shows what moved from the first draft to the film I delivered.

Part of the story	First draft	Final film
Setting and light	Rolling farm hills, with the light running from night through dawn to golden hour	Floating islands in bright daylight, with a calm sky
Main character	A farmhand in a straw hat	A girl with a wand, a hovering basket and a bee
Processing	Cherries pop out of a roaster as beans, then a hand grinder	A pulper, drying, roasting and grinding at the little mill
Finale	A giant cup used as a pool, with a duck and bean loungers	A pour into a giant cup, where a blossom opens
Title and brand	A bean mascot on a title card	A blossom logo, with the episode titles on posters

Part of the story	First draft	Final film
Palette	Ten colours from night indigo to cream	The same bright film colours (Table 2), with espresso, cream, kraft and ember for the brand

Table 3. From the first-draft storyboard to the final film.

Website storyboard

I ideated the website with Claude in chat before it was built. The idea moved in three steps, summarised in the table below.

Stage	What was decided
Early concept (September)	A specialty coffee site shown as a "how it's made" journey through the brewing stages, under the name BLOOM, From Cherry to Cup.
Landing idea (October)	A big, animated coffee cup on the landing page. Clicking it opens a Netflix-style episode section with a poster card for each episode.
Final structure	Three 40-second episodes with three posters, plus the pages listed in the next table: Home, Episodes, Meet the girl, How it's made and Behind the scenes.

Table 4. How the website idea developed during ideation with Claude.

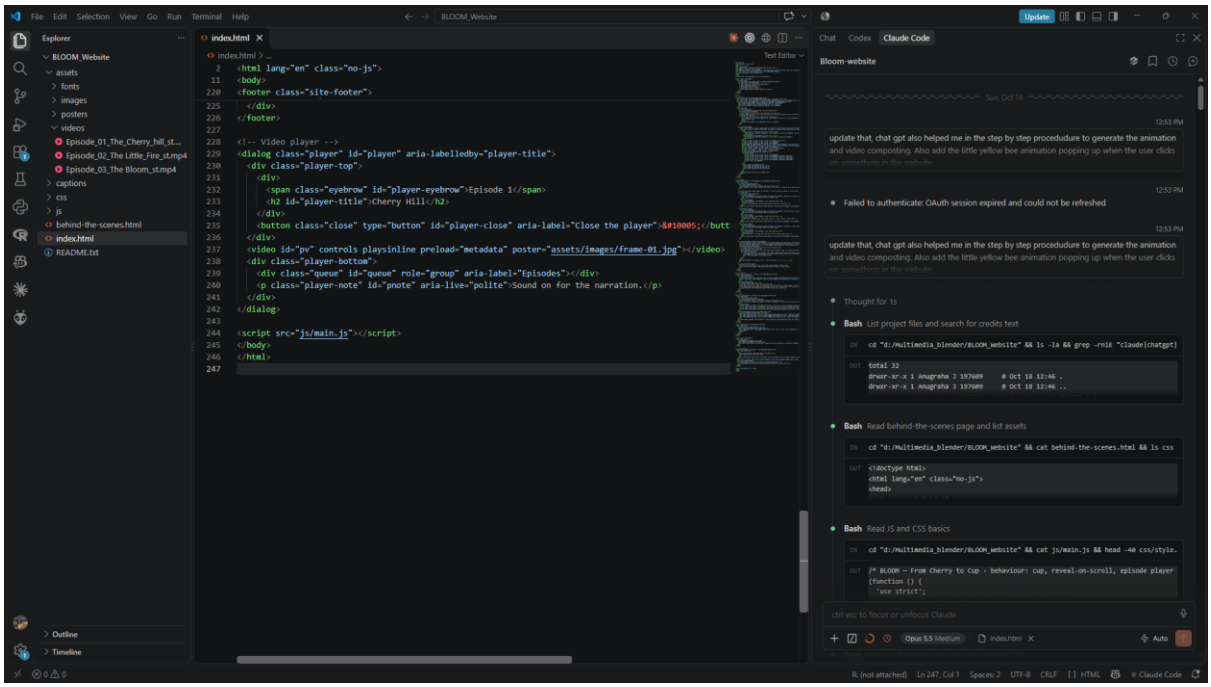


Figure 5. Screenshot of the ideation chat with Claude in VSCode

Page or part	Content	Purpose
Header	BLOOM wordmark and navigation: Home, Episodes, Meet the girl, How it's made, Behind the scenes	Constant navigation and clear location

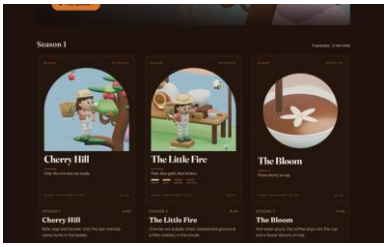
Page or part	Content	Purpose
Home	Large BLOOM title, a steaming cup you can click, and buttons to watch the whole film or choose an episode	A playful start that leads to the films
Episodes	Three poster cards with episode titles and a pop-up video player with captions	Choose and watch an episode, then return
Meet the girl	The character render and a short introduction	Connect the viewer to the character
How it's made	A short summary of the tools and steps	Show the production process
Behind the scenes	Who made what, own work or AI, plus tool credits	Transparent credits
Footer	Project name and unit	Consistent closing on every page

Table 5. Website pages and parts.

The storyboard became the interface below. Each panel is a screenshot of the website I built from it: the landing page with the clickable cup, the episode cards, and the pop-up player.



1. Home: click the cup



2. Episodes: choose a poster



3. Player: watch with captions

Figure 6. The website storyboard realised as the built interface.

Film storyboard

The film has three scenes of 40 seconds. Each runs 960 frames at 24 fps. The table gives the Scene 1 camera sequence, which matches the seven timeline markers in the Blender file (Section 3.3).

Frames	Time	Story action
1–96	0:00–0:04	Dawn and floating islands
97–216	0:04–0:09	Cart ride and balancing
217–336	0:09–0:14	Notice the bee and prepare
337–480	0:14–0:20	Crouch, leap and absorb the landing
481–672	0:20–0:28	Wand gestures and cherry collection
673–840	0:28–0:35	React to the bee
841–960	0:35–0:40	Leap towards the mill

Table 6. Scene 1 film storyboard.

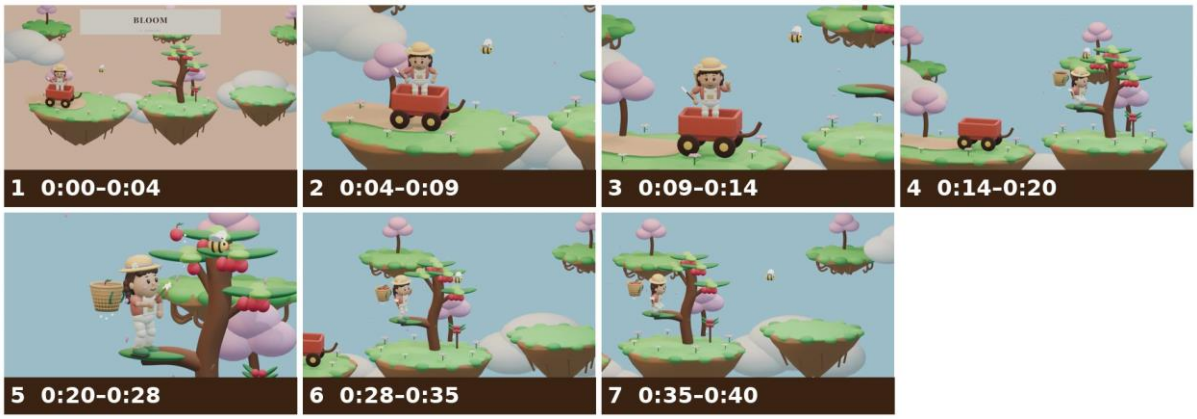


Figure 7. Storyboard image sequence for Scene 1, Cherry Hill. One frame per camera shot, taken from the episode video.

Scene 2, The Little Fire. The girl lands at the mill. Cherries enter the pulper, seeds emerge, drying time passes with a sun and moon cue, and beans roast and enter the grinder. She turns the crank, grounds fill a scoop, and she carries it towards the sky brewer. Drying and roasting use compressed story time.

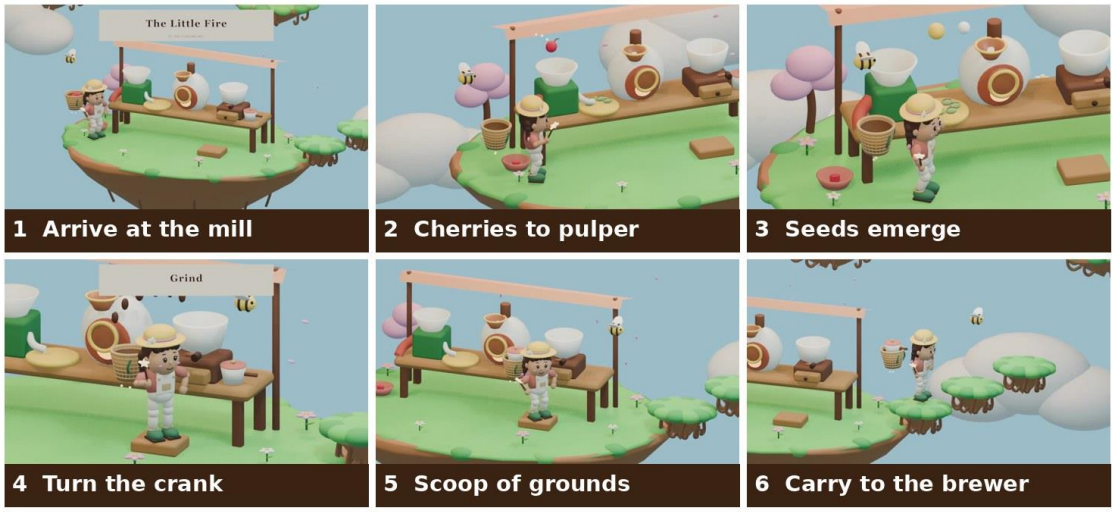


Figure 8. Storyboard image sequence for Scene 2, The Little Fire.

Scene 3, The Bloom. The scoop opens over the filter. The kettle tips and a stylised milk stream falls into it. A downward camera reveal shows a coffee waterfall reaching the giant cup. The coffee rises, a blossom opens, the girl waves and the final view shows the whole floating world.



Figure 9. Storyboard image sequence for Scene 3, The Bloom.

Audio and captions. Each scene has a short narration of seven or eight lines timed to the action, with music and sound effects underneath and captions on screen.

1.5 Response to Assignment 1 feedback

My Assignment 1 feedback said the topic was promising and the logo clearly reflected the coffee theme. It suggested three improvements: more detail in the banner to show the journey from cherry to cup, more readable poster text with a clearer visual hierarchy, and more 3D models, animation and video to tell the complete story. The points below explain how I responded to each suggestion.

- **Logo.** The feedback was positive, so I kept the coffee theme. As the story grew into a floating world, I re-ideated the coffee-bean mascot as a blossom, so the logo now matches the ending of the film, where a flower blooms on the coffee (Section 2.1).
- **Banner.** I redesigned the landing banner around the journey. It now shows a steaming cup with a blossom on the coffee over layered green hills, and leads into the three episodes. Each episode card and poster carries a three-step tracker (Harvest, Roast & Grind, Brew), and the Now showing feature steps through the episodes in order, so visitors can follow the journey from cherry to cup (Sections 2.2 and 6).
- **Posters.** I rebuilt the posters to improve readability and visual hierarchy, using the design principles from the lectures. Each poster now has one dominant title, a short subtitle and a tracker that highlights the current stage in an accent colour. A colour fade from the render into the espresso background keeps the text readable against the image (Section 2.3).
- **3D story and video.** I could not film myself as suggested, so I told the complete story in 3D with an animated character standing in for me. I modelled a cherry tree for the harvest, a roastery in the clouds as the processing stage, and a giant cup for the brew. These became three connected 40-second episodes, each narrated in my own voice. Instead of combining live footage with 3D, I separated the character and the environment into render layers and joined them with Alpha Over compositing (Sections 3 and 4). This kept every scene in one consistent visual style, and the play-all option joins the episodes into one film.

2 Image Enhancement

2.1 Logo

Design idea. The logo is a five-petal flower with an orange centre on a dark espresso rounded square. It repeats the blossom that opens on the coffee in the last scene, so the logo and the story end on the same image. It is simple enough to read at tiny sizes and is used as the website icon, on the posters and in the presentation.

How I made it. I designed and built the logo myself in Figma; Claude helped only with ideation. It has named layers (Background, Petals, Centre), and I exported it for the website, posters and slides. The petals are one shape repeated at 72-degree steps, which keeps the flower even and easy to edit. My Assignment 1 proposal used a coffee-bean mascot as the logo (visible in the top left of Figure 11). When the story grew into the floating world, I re-ideated the logo as a blossom to match the ending.



Figure 10. The BLOOM logo.

2.2 Banner

Assignment 1 version. The banner I proposed in Assignment 1 was a light cream landing page with the coffee-bean mascot, the headline "From cherry to cup.", a short introduction, an "Explore the journey" button and a navigation bar (About, The journey, Poster, Video, Gallery).

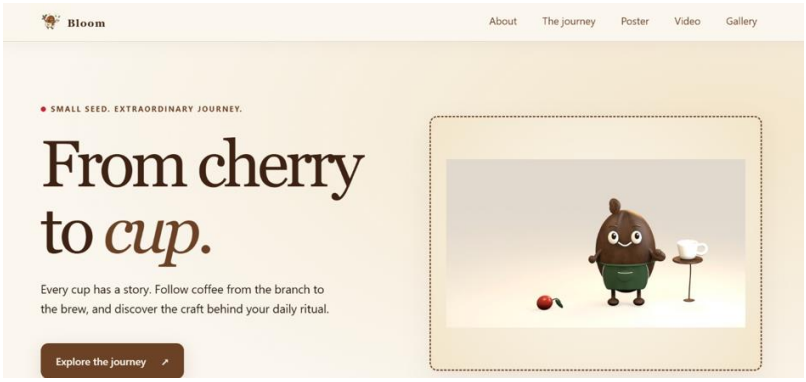


Figure 11. The Assignment 1 website banner, before it was enhanced.

Enhanced and re-ideated version. I enhanced and re-ideated the banner as the project grew into the floating-world story. The new landing area has a dark espresso background, a large BLOOM wordmark, a steaming cup with a blossom on the coffee, and layered green hills. It matches the logo and posters, sets the mood straight away and invites the visitor to click the cup. I coded it into the site myself, so it adapts to different screen sizes.



Figure 12. The enhanced website landing banner.

2.3 Poster

Design idea. There is one poster per episode. Each uses a full-bleed frame from my own Blender renders, the episode number and title, and a three-step tracker (Harvest, Roast & Grind, Brew) with the current stage highlighted in an accent colour. Together the three posters show the journey from cherry to cup. The titles are Cherry Hill, The Little Fire and The Bloom.

How I made them. I designed and built all three posters myself in Figma; Claude helped only with ideation. The hero images are my Blender renders. In Figma I handled the image crops, the colour fade from the render into the espresso background so the text stays readable, the typography and the tracker. These steps combine image cropping and scaling, tonal blending, and layered vector text over a render.

Strengths and weaknesses. The posters share one layout, so they read as a set, and each uses a scene-specific accent colour. The Episode 3 image is enlarged from a 960 by 540 preview, so it is softer than the others. Re-rendering that frame at 1920 by 1080 would fix this.



Figure 13. The three episode posters.

2.4 Other images

The other images are stills and frames rendered in Blender: the character, the island scenes and the layer renders used for compositing (see the figures in Sections 1, 3 and 4). I also made website screenshots to show the interface. The transparent character render is a working asset for compositing, not a finished poster.

3 3D Animation

3.1 Design idea

The 3D animation is the heart of BLOOM. It tells the coffee story through one consistent character in a miniature world. The film runs for two minutes, as three scenes of 40 seconds, 960 frames each at 24 fps. Each scene is set up for a 1920 by 1080 render and uses seven camera shots that alternate wide views with close-ups.

3.2 How I made it

I made the entire 3D animation in Blender. I followed step-by-step instructions given by ChatGPT, and I worked through each stage inside Blender myself. The same steps are written up in the BLOOM Beginner Teaching Workbook and the Blender Recreation Guide, which I include in my materials archive to show how each part is built.

1. **Materials and palette.** Named materials for every colour so related surfaces stay consistent.
2. **Environment.** Floating islands, trees, clouds, a cart and a bee built from editable meshes and curves.
3. **Character.** A girl with separate upper and lower limbs, hat, braids, hands, boots, a wand and a basket, built from parented objects.
4. **Props.** The grinder, mill, scoop, kettle, filter, funnel and giant cup.
5. **Animation.** Keyframed movement for the character, props and cameras.
6. **Cameras and lighting.** Seven shots per scene, then lighting and colour matched across scenes.
7. **Render.** Frames rendered from Blender and passed on for editing.

Cameras, lighting and rendering in detail. Each scene uses seven orthographic cameras, which give the miniature diorama look. Each camera is bound to a timeline marker at the start of its shot (Marker > Bind Camera to Markers), so Blender switches cameras automatically. Gentle camera moves are made by keying Location, Rotation and Orthographic Scale at the start and end of a shot; a lower scale zooms in. The scenes are lit with three area lights. The key light sits at (-7, -9, 15) as a warm 1920 W disk of size 9 (#FFE3B4), a 1040 W fill light at (7, -3, 10) softens the shadows, and a 1600 W rim light at (0, 8, 12) separates the subjects from the background. The full steps are in Appendix B. Each scene renders at 1920 by 1080, 24 fps, frames 1 to 960, as a numbered PNG sequence. Scene 1 uses Cycles for the layer separation in Section 4, and Scenes 2 and 3 use EEVEE. I turned each sequence into an H.264 MPEG-4 clip in Blender's Video Editing workspace before editing it in Canva.

3.3 Extra capabilities: Bezier curves and procedural animation

Beyond the basic lab techniques, I used Bezier curves and procedural animation with Python scripts. These concepts were taught in the lecture slides, and I took my inspiration from there. I applied them as follows.

Technique	Where it appears	What it does
Bezier curves	Milk stream, coffee waterfall, hanging roots, stems and branches	Smooth, editable shapes that can be bevelled into tubes
Curve animation	Milk stream and waterfall	Points, handles and thickness are animated so the stream grows and follows the kettle and coffee level
Scripted limb poses	Knees, elbows and grinder grip	A script calculates the middle joint from the two end points and keys the result
Staggered timing	Cherries, seeds and grounds	The same action repeats with small time offsets
Repeating cycles	Wings, clouds and petals	A small keyed pattern repeats using curve modifiers

Table 7. Procedural techniques used in the animation.

The Animation workspace in my Scene 1 file shows how this looks in practice. The Graph Editor holds dense, regular curves for the cherry, the Dope Sheet shows its keys and the keyed sky, and the timeline markers 01 Dawn islands to 07 Leap to the mill match the film storyboard.

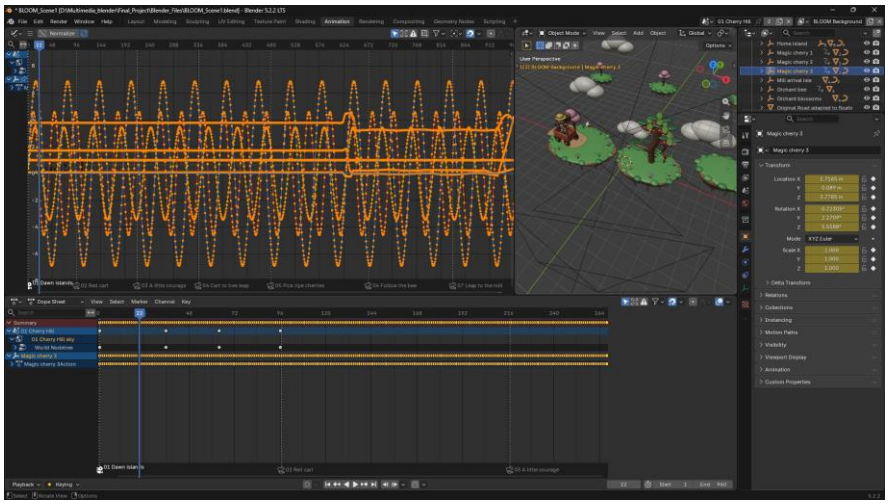


Figure 14. Blender Animation workspace in BLOOM_Scene1.blend: Graph Editor (top left), 3D viewport (top right) and Dope Sheet (bottom) for the object Magic cherry 3, with markers 01 to 07 and the frame range 1 to 960.

The streams are stylised procedural animation, not a fluid simulation. Their strength is direct control and predictable rendering with no simulation cache. Their limitation is simplified physical behaviour.

How the milk stream is built. The stream is a 3D Bezier curve with a bevel depth of 0.055, which turns the line into a smooth tube, and fill caps close its ends. The pour is animated on the curve

data, not the object. The bevel End factor is keyed from 0 at frame 300 to 1 at frame 314, so the head of the stream travels from the spout to the filter. The Start factor is keyed from 0 to 1 between frames 851 and 865, so the tail follows and the pour ends. In the final file the curve handles are keyed on every frame, so the ends stay attached to the kettle spout and the filter as the kettle moves.

How the limb script works. For each knee or elbow, the script takes the distance d between the two end joints, such as the shoulder and the hand, and the two segment lengths L1 and L2. The middle joint lies $a = (L1^2 - L2^2 + d^2) / (2d)$ along that line and is offset sideways by $\sqrt{(L1^2 - a^2)}$. A chosen bend direction puts it on the correct side. The script then places each visible segment between its joints and keys the result. This is a simple two-segment inverse kinematics solve, so I could pose limbs from hand and foot positions instead of rotating each part by hand.

3.4 Character construction and movement

The girl is built from parented native objects and Empty controls. The top-level control is named Girl - motion root. It is an object hierarchy with keyed controls, not a skinned bone armature. The Scene 1 hierarchy has 137 objects, and clear names such as Girl Left forearm make them easy to find in the Outliner and Graph Editor.

The jump has planted feet during preparation, a crouch, a push-off, a leg tuck, landing compression and recovery. Small offsets on the head, hat, braids and basket create follow-through. The cart ride includes balance gestures. Three cherries leave the tree, follow arcs above the girl and land in the basket. At the mill her hand follows the grinder crank so the action reads as contact.

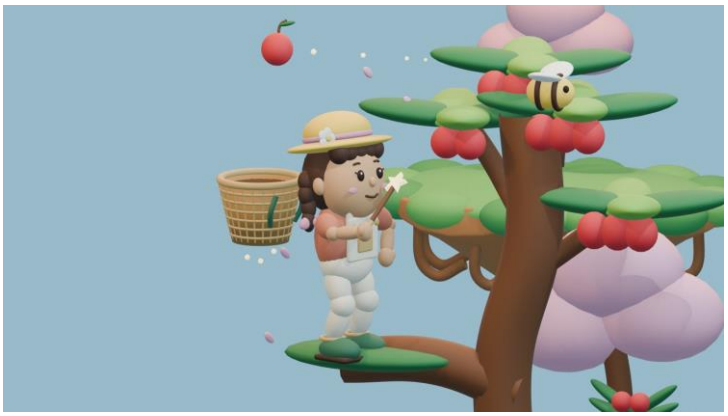


Figure 15. The character, wand, basket and cherry tree in Scene 1.

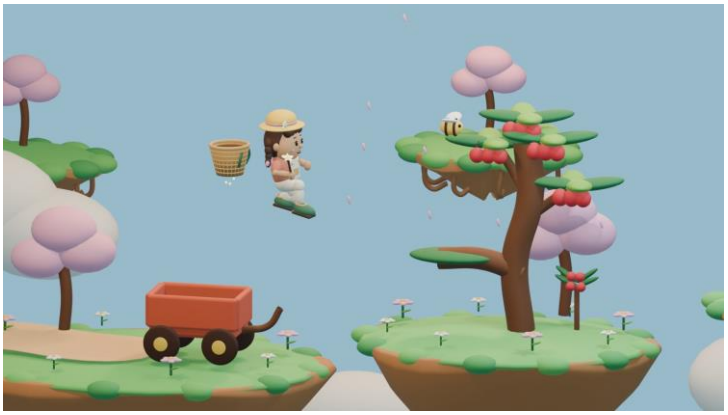


Figure 16. A jump pose used to review the character animation.

3.5 The wave: FreeMoCap attempt and Blender animation

I wanted the girl's wave to come from my own movement, so I tried to capture it with FreeMoCap. My laptop was not able to process the recording. It repeatedly crashed or took a very long time. Because I could not finish that workflow, I animated the wave directly in Blender with keyframes, using the Animation workspace and the Graph Editor. The captured motion was not used on the character. The result can be inspected in my Blender files, where the keyed wave appears in the Dope Sheet and Graph Editor, as in the workspace shown in Section 3.3. The trial taught me to choose a workflow that is practical on the available hardware.

3.6 Evaluation

The strengths are a consistent character, editable assets and controlled procedural effects. The limits are that some previews are 960 by 540, the streams simplify real fluid behaviour, and the per-frame keys make dense animation curves that are harder to edit by hand.

What I learned and would do differently. Building BLOOM taught me how parent hierarchies and clear names keep a 137-object character manageable, the difference between keying curve data and keying object transforms, and how a short script can solve limb poses. The FreeMoCap trial taught me to choose a workflow that suits my hardware. Next time I would rig the girl with a bone armature and IK constraints, so the animation curves stay light and easy to edit. I would also render every scene at full resolution, and, now that I know the Alpha Over workflow, try my tutor's suggestion of filming live footage and compositing it with the 3D world.

4 Video Composition

4.1 Design idea

I wanted the girl to be a separate layer so she could be placed over the environment with full control. I explored this out of pure interest, as a technique beyond the basics. It is built in the Blender compositor for Scene 1, and the full composited version is used in the final Scene 1 episode.

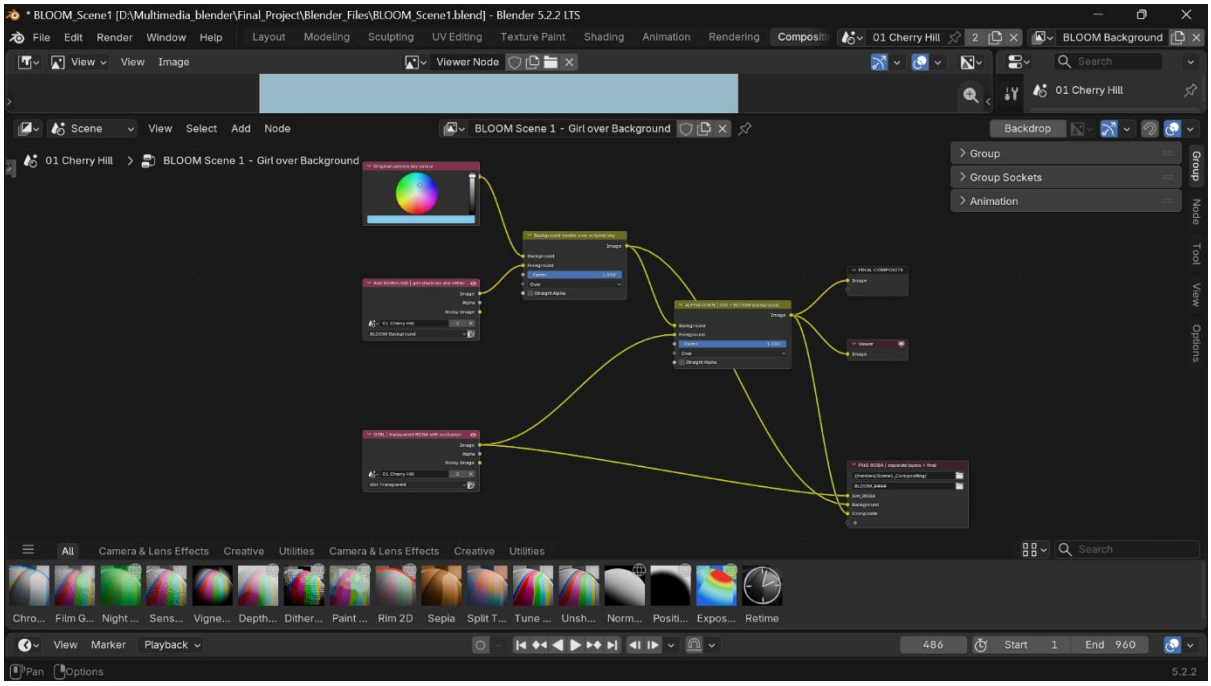


Figure 17. The Scene 1 compositor node tree in Blender, with the girl layer placed over the background by an Alpha Over node.

4.2 Alpha Over composition

Scene 1 has an active, connected compositing setup. In the 01 Cherry Hill scene I confirmed the features in the table below.

Implemented feature	What it does
BLOOM Background render layer	Separates the scenery while retaining the girl's shadows and reflections through Indirect Only.
Girl Transparent render layer	Renders the girl with transparency. Background Holdouts preserve objects appearing in front of her.
Restore BLOOM Sky (Alpha Over)	Places the background render over the original sky colour.
Girl over BLOOM Background (Alpha Over)	Combines the transparent girl with the completed background.
Final Composite, Viewer and File Output	Receives the combined image and provides outputs for the girl, background and composite.

Table 8. Confirmed compositing features in Scene 1 (01 Cherry Hill).

Compositing is enabled, and Scene 1 is configured for Cycles at 1920 by 1080, frames 1 to 960 at 24 fps. The combined film timeline also references Scene 1.

The first Alpha Over node restores the original sky behind the environment. The second places the transparent girl over that complete background. Its factor is one. In this way I applied a video compositing technique on top of the animation. The table below gives the setup in more detail.

Stage	Implementation
Collections	BLOOM Girl holds all 137 character objects. BLOOM Background holds the environment.
Render settings	Cycles CPU, 64 samples, denoising, adaptive sampling, Film Transparent.
Background view layer	The girl collection is Indirect Only, so her shadow and light are kept while she is hidden.
Girl view layer	The environment collection is Holdout, so objects in front of the girl still cover her.
Sky restoration	A colour matching the original camera sky feeds the first Alpha Over node.
Character combination	Girl Transparent is the foreground and the restored environment is the background of the second Alpha Over node. Factor is 1.
Outputs	Final output, Viewer, and separate Girl, Background and Composite PNG outputs.

Table 9. The Scene 1 compositing setup in detail.

Alpha Over uses the foreground alpha channel to combine the images. Blender render passes use premultiplied alpha, so Straight Alpha stays off. Both passes use the same camera, so they stay aligned.

Reproducing the setup. In Render Properties, choose Cycles with 64 samples, denoising and adaptive sampling (noise threshold 0.025), and enable Film > Transparent. Move the girl into a BLOOM Girl collection and the environment into BLOOM Background. In the first view layer, set BLOOM Girl to Indirect Only; in a second view layer, Girl Transparent, set BLOOM Background to Holdout. In the Compositing workspace, connect the sky RGB node to the Background input of the first Alpha Over node and the background render to its Foreground. Connect that result to the Background of the second Alpha Over node and the Girl Transparent render to its Foreground, with both factors at 1. The second node feeds the Group Output and the Viewer, and a File Output node saves Girl_RGBA, Background and Composite as 8-bit RGBA PNG images.



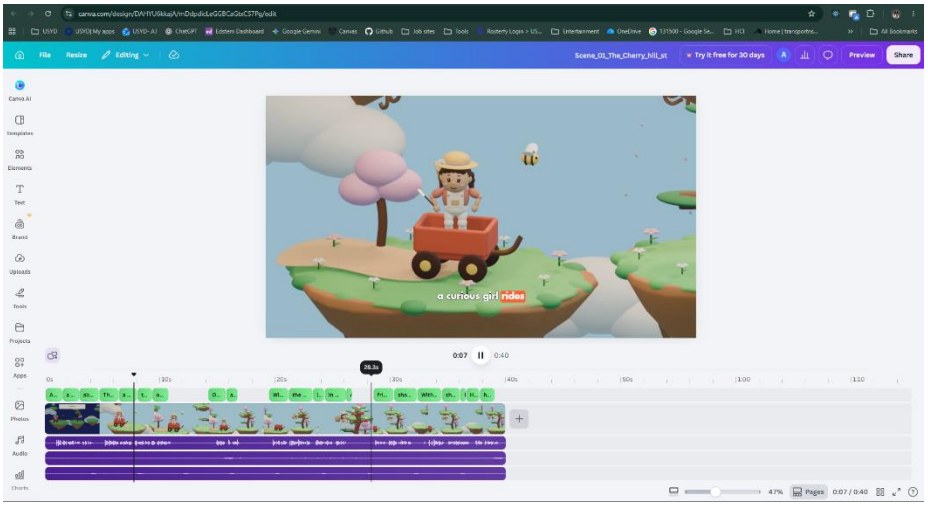
1. Background 2. Girl, transparent 3. Composite

Figure 18. The two render layers and the Alpha Over result at frame 280, verified at 1920 by 1080.

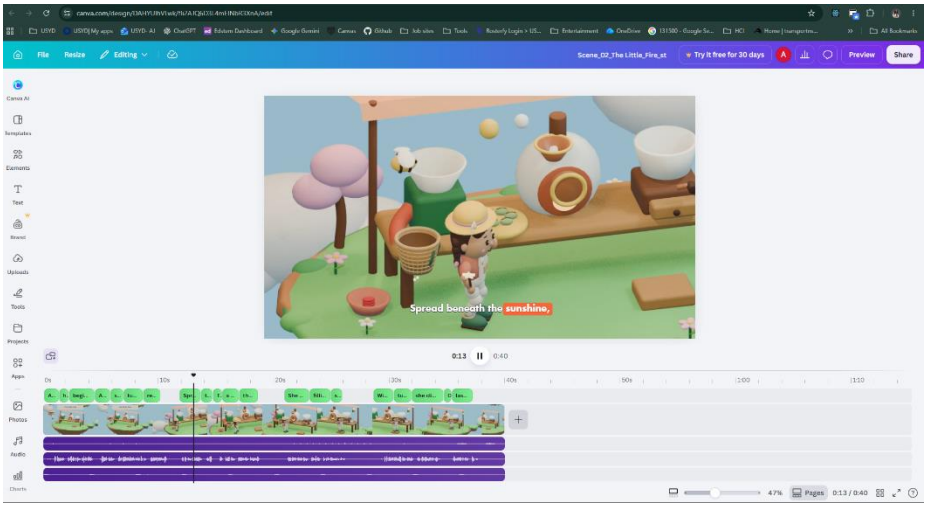
A full-resolution example at frame 280 is verified, with separate girl, background and composite images. The full composited sequence was rendered and is the version used in the final Scene 1 episode.

4.3 Editing and synchronising in Canva

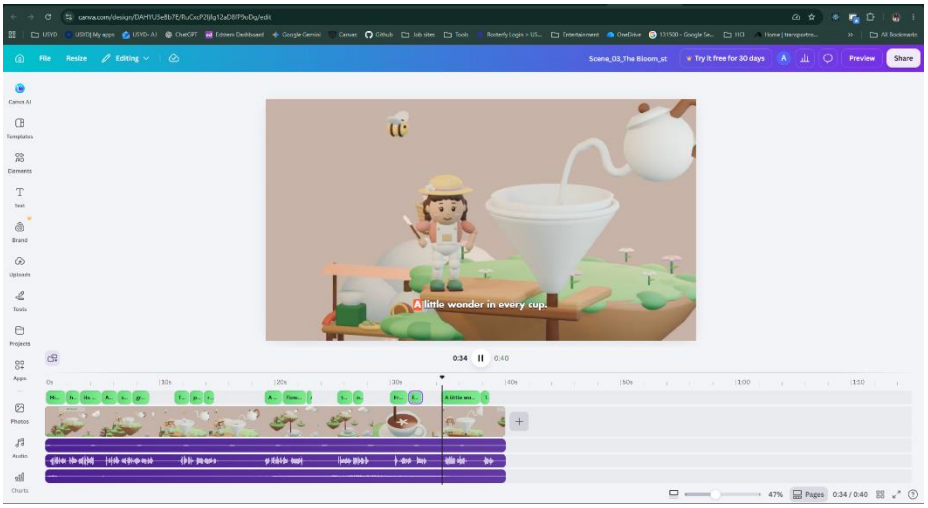
I used Canva to edit the episodes, sync the audio to the video, combine the narration with the music and sound effects, and add subtitles. This also kept the narration lines in step with the action on screen. The finished episodes are exported as MP4 files for the website. These three 40-second episodes are the project's video trailers: each one introduces a stage of the journey from cherry to cup with narration, music, sound effects and subtitles, and together they tell the whole story. The screenshots in Figure 19 show my Canva timelines for the three episodes, where I placed and timed the subtitle blocks and lined up the narration, music and effects tracks against the video myself.



(a) Scene 1, Cherry Hill



(b) Scene 2, The Little Fire



(c) Scene 3, The Bloom

Figure 19. My Canva editing timelines for the three 40-second episodes, showing the video clip, the three audio tracks (narration, music and effects) and the subtitle blocks that I synced by hand.

5 Audio

5.1 Design idea

The audio is meant to give the film the warm feel of a storyteller. The narration is in my own voice, with its pitch altered so it sounds like a narrator, and it supports the toy-like world, while music and effects mark contacts and transformations, such as the cherries landing, the grinder turning and the blossom opening.

5.2 How I made it

- **Script.** I wrote the narration script myself. It has seven or eight short lines per scene, each matched to an action.
- **Voice.** The voice is my own. I recorded it, then used ChatGPT to alter its pitch so it sounds more like a narrator. Only the pitch was changed; the voice is still mine.
- **Copyright.** Because the voice is mine, there are no third-party voice copyright issues.
- **Sync and subtitles.** I edited and synced the audio to the video in Canva, and added the subtitles, timing each caption to the narration by hand.
- **Music and effects.** I fetched the music and sound effects using ChatGPT. I then combined them with the narration and the video myself in Canva, so the final mix is my own assembly.

Music and Effects	Verified source and method
Background music	Python/NumPy synthesis using programmed melodies, chords, sine waves and harmonics.
Sound effects	Synthesized whooshes, pops, grinder sounds, pouring/water sounds and magic chimes, using noise, tones and volume envelopes.

Music and Effects	Verified source and method
External recordings	The inspected music/effects scripts do not import stock recordings or downloaded samples.

Table 10. Sources of the music and sound effects.

5.3 Evaluation

The narrated voice gives the story warmth and a clear thread, and the captions help viewers who cannot listen. Pitch-shifting can make a voice sound slightly processed, so I reviewed the balance between voice, music and effects by listening in Canva.

6 Web Interface

6.1 How the website fits the project goal

The website turns the three films into one experience. It opens with a playful landing, where a click on the cup leads to the episodes, and each episode plays in a pop-up player. It carries the same dark espresso theme, logo and posters as the rest of the project. The home page presents the three episodes as poster cards (Figure 20) and features one episode at a time, with an option to play all three in order (Figure 21). Figure 22 shows the pop-up player.

The content is my own: the text, the films, the posters and the credits. I made the website myself in HTML, CSS and JavaScript. The ideation was assisted by Claude, and I later enhanced the site using Claude Code: the website animations and bee graphics were made with Claude's assistance, and Claude helped me push the code to GitHub and deploy it. The site is built locally with relative file paths, so it opens from the folder or from a simple local server. The finished site is also deployed on GitHub Pages at https://anu-aj.github.io/bloom-cherry-to-cup/index.html.

6.2 Interactive navigation and usability

- A visible navigation bar with the current page marked, and a skip link for keyboard users.
- Poster cards open a video player with controls for play, pause, seek and mute. The player closes with a button or the Escape key and returns the visitor to the episode list.
- Sound is streamed with the video. Each episode's narration, music and effects are mixed into its MP4 file, which the browser streams through the HTML5 video player, so playback and sound start while the file is still loading.
- A play-all option runs the three episodes in order.
- Captions are available on every episode.
- Layouts, images and videos scale to different screen sizes, including narrow phone screens.
- Animations respect the visitor's reduced-motion setting, and images have text descriptions.
- The Behind the scenes page credits the AI used and the tools used, so visitors can see who made what.

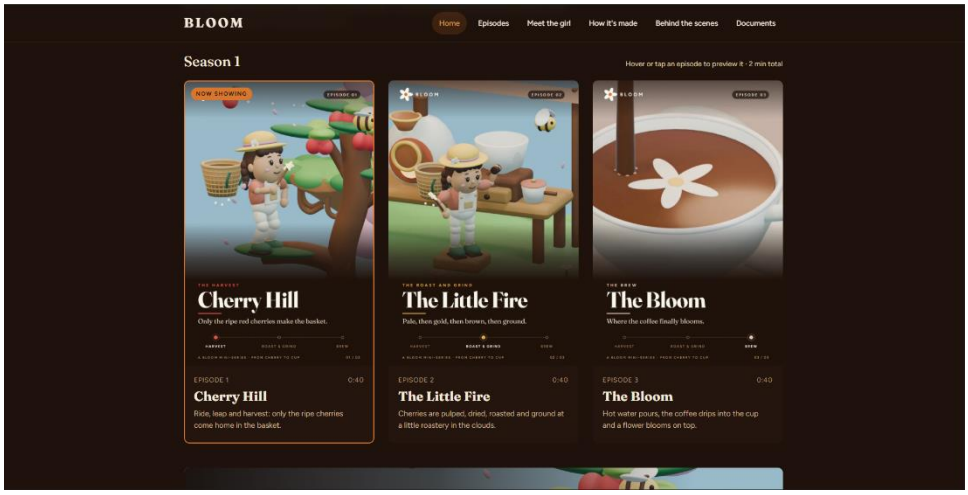


Figure 20. The Season 1 episode cards on the home page. Each card reuses its episode poster, and hovering or tapping a card previews the episode.

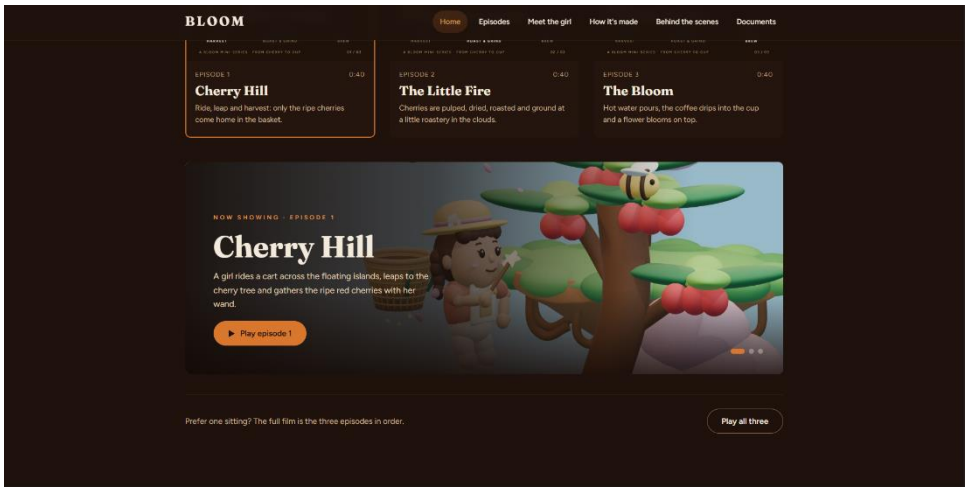


Figure 21. The Now showing feature for Episode 1, with its Play episode button, and the Play all three option for watching the full film.

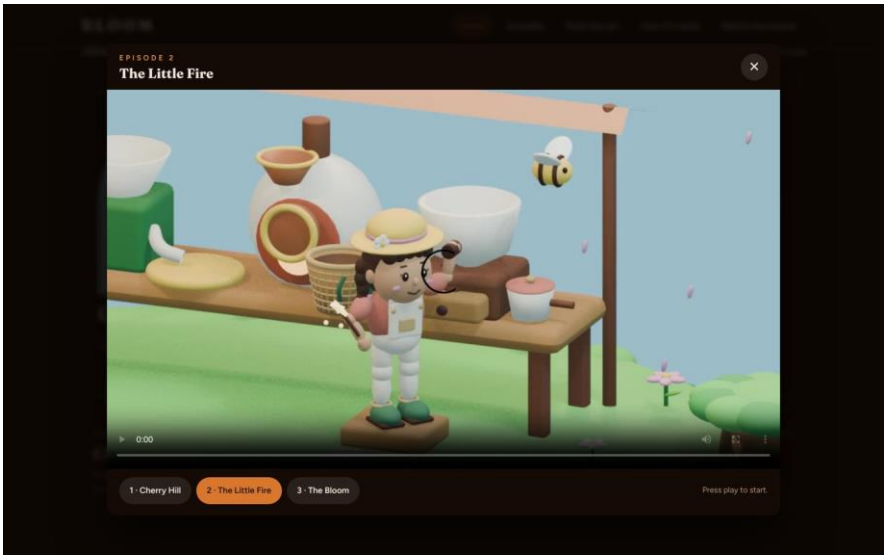


Figure 22. The pop-up episode player, with buttons to switch between the three episodes.

6.3 Testing and known limitations

Testing: I manually tested the website, to verify if all the sections are working as expected and then checked if the video quality is rendered as expected. Table 11 lists the tests I ran on the finished production.

Test	How I checked it	Expected result	Result
Navigation	Clicked every link in the navigation bar on each page	Each link opens the right page and the current page is marked	Pass
Episode player	Opened each poster card and used play, pause, seek and mute	The episode plays with its sound, and every control works	Pass
Play all	Started the play-all option	The three episodes play in order	Pass
Captions	Watched each episode with captions on	Captions appear on screen in time with the narration	Pass
Video quality	Watched each episode in the website player	Renders, compositing and text display as expected	Pass
Keyboard access	Used Tab, the skip link and the Escape key	All controls can be reached, and Escape closes the player	Pass
Screen sizes	Resized the browser down to phone width	I Did not make website responsive; it is made only to be viewed in desktop. So, it should not be responsive	Pass.
Behind the scenes page	Opened the page from the navigation bar	AI use and tools are credited	Pass

Table 11. Tests run on the finished website and episodes.

Known limitations: some previews are 960 by 540, and video files are large, so first playback depends on the local disk or server.

7 AI acknowledgement and reflection

I used two AI tools in this project: ChatGPT and Claude, including Claude Code, which helped me enhance the website I had made, assisted with its animations, and helped me push the code to GitHub and deploy it. I have kept the full prompts in my AI acknowledgement form, and the website's Behind the scenes page credits the AI and the tools used so visitors can see who made what. A screenshot of the website ideation chat is in Figure 5, and the BLOOM Beginner Teaching Workbook and Blender Recreation Guide in my materials archive show how each Blender stage was built. Claude assistance was also used to write this document. An earlier draft was prepared with ChatGPT, and Claude helped me rewrite and edit it from my own account of the work, and assisted with the figure captions, the contents and lists, and this acknowledgement section. The table

separates what each AI did from what I did. Figures 23 and 24 show examples of ChatGPT's assistance: its explanation of the Python scripts and procedural modelling methods, and of the parenting hierarchy for the girl character.

Tool	What it was used for	My contribution
ChatGPT	Step-by-step Blender instructions for modelling and animation, Python scripts for Bezier and procedural animation, compositor guidance for the Alpha Over setup, fixing overlaps and errors, altering the pitch of my recorded voice, fetching the music and sound effects (made with Python/NumPy synthesis, not stock recordings), and an earlier draft of what should be included in this report	Followed the instructions in Blender, built and animated the scenes, keyframed the wave myself, set up and explored the compositing, wrote the script, recorded my voice, mixed the music and effects with the video in Canva and checked the results
Claude	Ideation for the first-draft storyboard, logo, posters and website; through Claude Code, enhancing the website after I had made it, assisting with the website animations and bee graphics, and helping me push the code to GitHub and deploy it; the presentation structure; writing this report, including rewriting the earlier draft from my account of the work, editing, figure captions, the contents and lists, and this acknowledgement section	Directed the ideas, designed the first-draft storyboard in Figma, designed and made the logo and the posters in Figma from my own Blender renders (crops, colour fades and typography), supplied the content, media and credits, made the website in HTML, CSS and JavaScript, then directed and checked the enhancements and the deployment, reworked and finished the slides in Canva, and supplied my own account of the work for this report and checked the final text
Figma, Canva, Blender, FreeMoCap	Design, editing and animation tools. FreeMoCap was tried for the wave, but my laptop could not process the recording, so it was not used in the end	Used directly by me
Pinterest and Google Images	Reference images for the mood board only (Figure 2)	Used directly by me from the sources.

Table 12. What each AI tool was used for, and my own contribution.

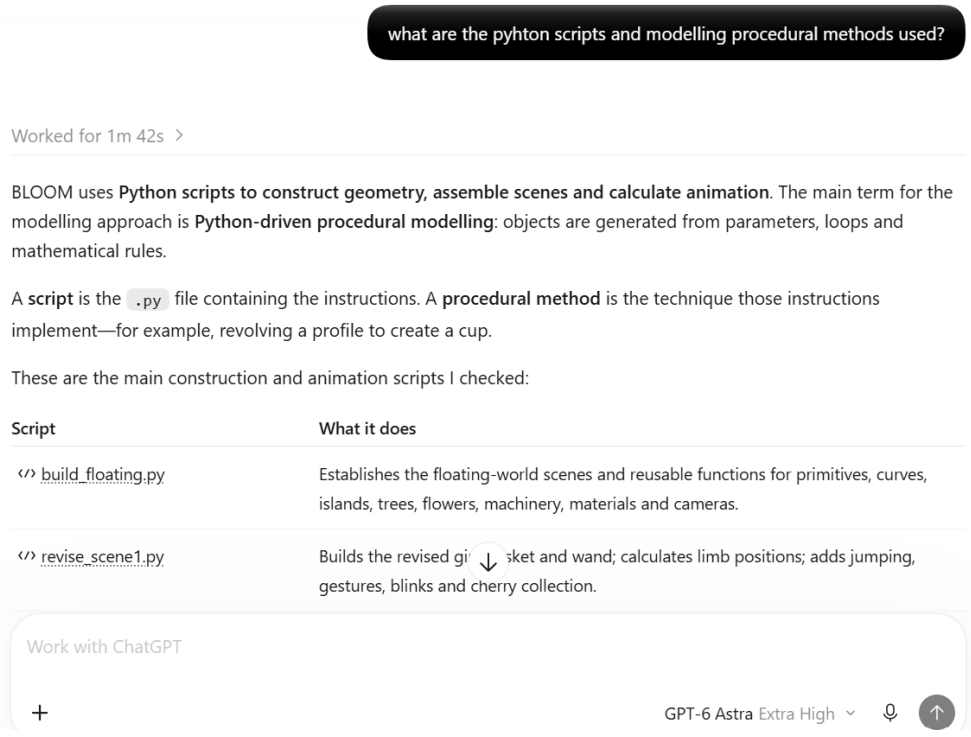


Figure 23. ChatGPT explanation of the Python scripts and procedural modelling methods used in BLOOM.

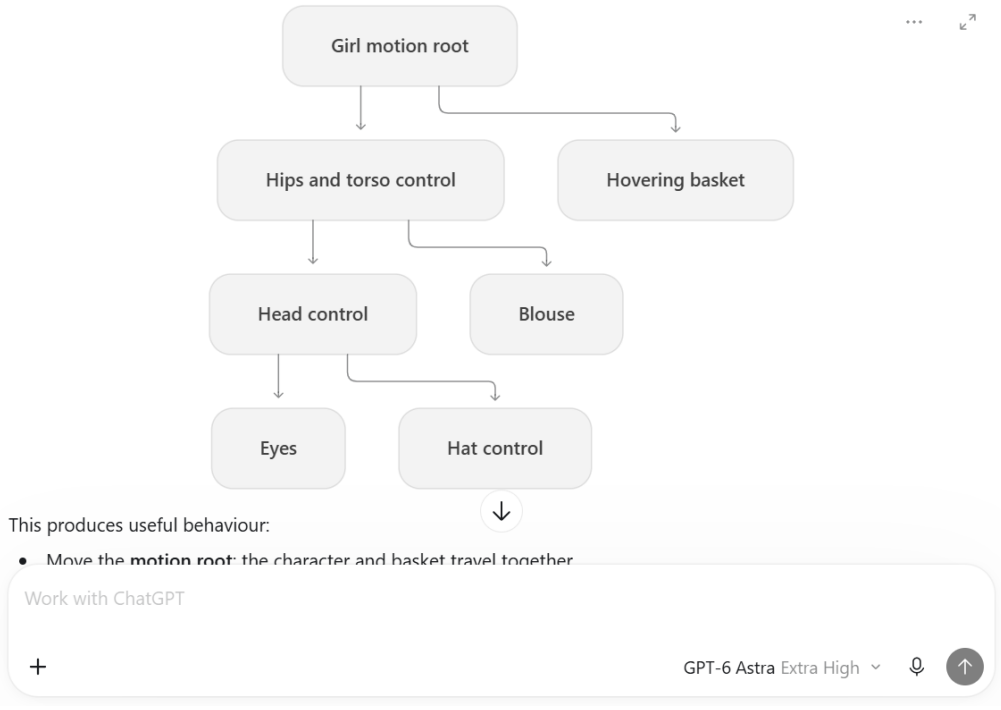


Figure 24. ChatGPT explanation of the parenting hierarchy used for the girl character.

Work done without AI

These parts of BLOOM were made by me without AI help:

- The three-episode story and the narration script.
- The narration voice, which I recorded myself. ChatGPT only altered its pitch.
- The mood board, gathered from reference images before building.
- The Blender scenes and character animation, including the wave, which I keyframed by hand after the FreeMoCap attempt.
- The final logo and poster artwork in Figma, built from my own Blender renders.
- The first version of the website, which I made in HTML, CSS and JavaScript. Its later enhancements and deployment are credited to Claude in the table above.
- The Canva edit: syncing the audio to the video, mixing the narration with the music and effects, and adding the subtitles, with each caption timed to the narration by hand.

Evaluation of the assistance

AI made repetitive scene construction faster, gave me step-by-step instructions I could follow, and helped me explore ideas for the logo, posters and website quickly. It also needed checking. An animated scene is not automatically video compositing, and a motion-capture trial is not automatically usable character motion. Some suggestions needed correcting in Blender, and the FreeMoCap route had to be replaced with my own keyframes.

The procedural approach gives editable assets but also dense curves and stylised motion. Separating layers improves compositing control, while Cycles can change the earlier Eevee look and takes longer to render. The story also changed from the early farm storyboard, so this report describes the version I implemented.

The final submission is my own work. The ideas, the Blender build, the voice, the Figma artwork and the account of the work in this report are mine, and the AI contributions are named above so they can be assessed separately.

Reference

[1] COMP5415/4415 Multimedia Design and Authoring. Assignment 2: Multimedia Production Implementation and Delivery. Brief, 2026.

[2] COMP5415/4415 Multimedia Design and Authoring. Assignment 3: Presentation and Demonstration. Brief, 2026.

[3] COMP5415/4415 Week 03- 09 lecture slides on Bezier curves and procedural animation, which inspired the techniques in Section 3.3.

[4] Blender Manual. Alpha Over Node. https://docs.blender.org/manual/en/latest/compositing/types/color/alpha_over.html (accessed 18 October 2026).

[5] Blender Manual. View Layers Introduction. https://docs.blender.org/manual/en/4.5/render/layers/introduction.html (accessed 18 October 2026).

[6] BLOOM Beginner Teaching Workbook and BLOOM Blender Recreation Guide, project documents, October 2026.

[7] Fraunces and Figtree typefaces, Google Fonts, used on the website and posters under the SIL Open Font License.

[8] Assignment 1 website: The first ideation for this project – Old live website: https://anu-aj.github.io/Digital-Media/#home

[9] Project evidence: Blender scene files with embedded scripts, compositing build and verification records, narration script, caption files, Figma source files and the website folder. Live website: https://anu-aj.github.io/bloom-cherry-to-cup/index.html

[10] Mood board reference images (Figure 2) collected from Pinterest and Google Images, used for creative reference only and not reproduced in the final work.

Apart from the mood board reference images, all figures are renders, artwork or screenshots from the BLOOM project.

Appendix A First-draft storyboard and mood board

This is the full first-draft page, exported from Figma. It was the starting point for the colours, the shot list and the tone of the film.

Figma source file (logo, posters and first-draft storyboard): COMP5415_Bloom_Project_assets in Figma

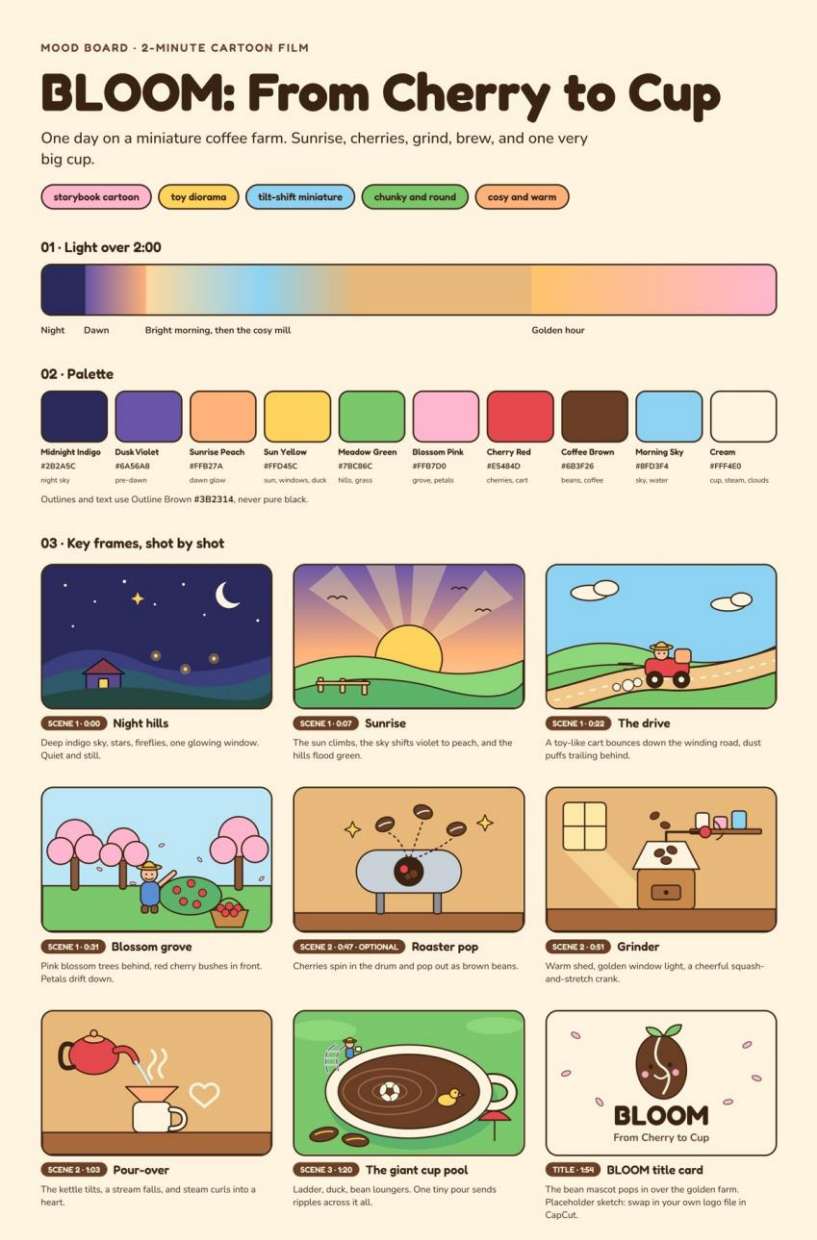


Figure 25. First-draft mood board and storyboard, exported from Figma.

Appendix B Blender reproduction steps

These steps summarise how the key Blender techniques in Sections 3 and 4 can be reproduced. They are condensed from the BLOOM Beginner Teaching Workbook in my materials archive, which gives every modelling and animation step in full.

B.1 Milk stream (Bezier curve animation)

1. Choose Add > Curve > Bezier, rename the object Milk_Stream, and set its Location and Rotation to (0, 0, 0).
2. In Edit Mode, place the first endpoint just inside the kettle spout and extrude further points (Control Points > Extrude Curve and Move) down to the filter.
3. Select all points and choose Control Points > Set Handle Type > Automatic, then return to Object Mode.
4. In Curve Data Properties, set Shape to 3D, Resolution Preview U to 32, Bevel Depth to 0.055 and Bevel Resolution to 5, enable Fill Caps, and assign the S3 silky milk material.
5. Keep the ends attached as the kettle moves. In the final file the curve handles are keyed on every frame; a simpler alternative is Hook to New Object on each endpoint, with the start Empty parented to Kettle_ROOT and the end Empty parented to Funnel_ROOT (Keep Transform).
6. Key Curve Data > Geometry > Bevel > End factor at 0 on frame 300 and 1 on frame 314, so the head of the stream travels to the filter.
7. Key the Start factor at 0 on frame 851 and 1 on frame 865, so the tail follows and the pour ends.
8. Play the timeline and check that the stream starts at the spout, ends in the filter and never shows a gap.

B.2 Cameras

1. Choose Add > Camera, name it after the shot (for example 01 Dawn islands) and set Type to Orthographic in Camera Data Properties.
2. Enter the camera's Location, Rotation and Orthographic Scale from Table 13.
3. Set the timeline to the shot's start frame and choose Marker > Add Marker.
4. Select the marker and the camera, then choose Marker > Bind Camera to Markers.
5. Repeat for all seven shots, and preview each with View > Cameras > Active Camera.
6. For a gentle move, key Location, Rotation and Orthographic Scale at the start of the shot, change them slightly at the end and key them again. A lower scale zooms in.
7. For a camera that must follow a subject, add a target Empty and a Track To constraint (Track −Z, Up Y), then animate the target.

Start frame	Camera	Location	Rotation (°)	Ortho scale
1	01 Dawn islands	(5, -20, 10)	(68.06, 0, 13.71)	19.0
97	02 Red cart	(-1, -14, 6)	(72.07, 0, 13.84)	9.0
217	03 A little courage	(0, -12, 5)	(76.10, 0, 10.94)	8.0
337	04 Cart to tree leap	(4, -18, 7)	(77.26, 0, 14.34)	12.0
481	05 Pick ripe cherries	(3.8, -8.5, 4.8)	(80.06, 0, 19.10)	5.7

Start frame	Camera	Location	Rotation (°)	Ortho scale
673	06 Follow the bee	(5, -15, 8)	(72.65, 0, 8.80)	12.0
841	07 Leap to the mill	(7, -18, 8)	(74.77, 0, 8.91)	12.0

Table 13. Scene 1 camera values at the start of each shot.

B.3 Lighting and colour management

1. Add an area light named Key_Light at (−7, −9, 15), rotation (41.25°, 0, −37.88°), 1920 W, Disk shape, size 9, colour #FFE3B4.
2. Add Fill_Light at (7, −3, 10), rotation (47.41°, 0, 66.80°), 1040 W, size 8, colour #DAF0FF.
3. Add Rim_Light at (0, 8, 12), rotation (41.63°, 0, 180°), 1600 W, size 7, colour #FFD0A0.
4. In Color Management, set View Transform to AgX, Look to Medium High Contrast, Exposure 0 and Gamma 1.
5. Check the result with Render > Render Image, because Material Preview lighting differs from the final render.

B.4 Alpha Over compositing (Scene 1)

1. In Render Properties, choose Cycles with 64 samples, denoising and adaptive sampling (noise threshold 0.025), and enable Film > Transparent.
2. Move Girl_ROOT and all its children into a BLOOM Girl collection and the environment into BLOOM Background. Keep cameras and lights in separate shared collections.
3. Rename the first view layer BLOOM Background and set the BLOOM Girl collection to Indirect Only in it.
4. Create a second view layer, Girl Transparent, and set BLOOM Background to Holdout in it.
5. In the Compositing workspace, add a Render Layers node for each view layer, an RGB node for the sky, two Alpha Over nodes, a Viewer, a Group Output and a File Output node.
6. Connect the sky RGB to the first Alpha Over Background and the BLOOM Background render to its Foreground.
7. Connect the first Alpha Over result to the second Alpha Over Background and the Girl Transparent render to its Foreground. Connect the second result to the Group Output and the Viewer.
8. Set both Alpha Over factors to 1 and leave Straight Alpha off, because the render layers are premultiplied.
9. Set the File Output to PNG, RGBA, 8-bit, with outputs Girl_RGBA, Background and Composite.
10. Render a still and check that the girl has a transparent surround, foreground objects still cover her, and the composite includes the environment and sky.

B.5 Rendering and export

1. In each scene, set the output to 1920 by 1080 at 100 percent, 24 fps, frames 1 to 960, as PNG (RGBA where transparency is needed), with a separate output folder per scene.
2. Render test frames at the jump, harvest, crank, pour and final wide shot, and fix any intersections or framing problems.

3. Choose Render > Render Animation. Each scene produces 960 numbered images.
4. In a new Video Editing file set to 24 fps, use Add > Image/Sequence to import a scene's images in order.
5. Set the output to FFmpeg Video, MPEG-4 container, H.264 codec, and render the clip. Repeat for the other scenes.
6. In Canva, place the three clips in orchard, processing and pour order, add the narration, music and effects, and export each episode.